1. a) 7C02H = 0111 1100 0000 0010

TI = 0 (GDT)

DI = 1111 1000 0000 = F80H

b) End = Base + Limit * 2K

= 1F308810H + 23BD77FFH

= 42EE000FH

c) Access: 54H = 0101 0100

A = 0, so access is not granted.

d) DPL = 10

e) E = 0, ED = 1. So, SS.

f) Total number of rows in GDT & LDT

$= 2^{13} + 2^{13} = 2^{14} = 16,384$

Limit = 20bits. So each descriptor represents a memory segment of size(maximum) = $2^{20} * 2KB$

∴ Max virtual memory size = $2^{14} * 2^{20} * 2KB$

$= 2^{45}$

= 32TB

= 2^15 GB

2. (a) Duty cycle = $\frac{4}{5}$ = 0.8 = 80%

100% D.C. $\Rightarrow 2^7 - 1 = 127$

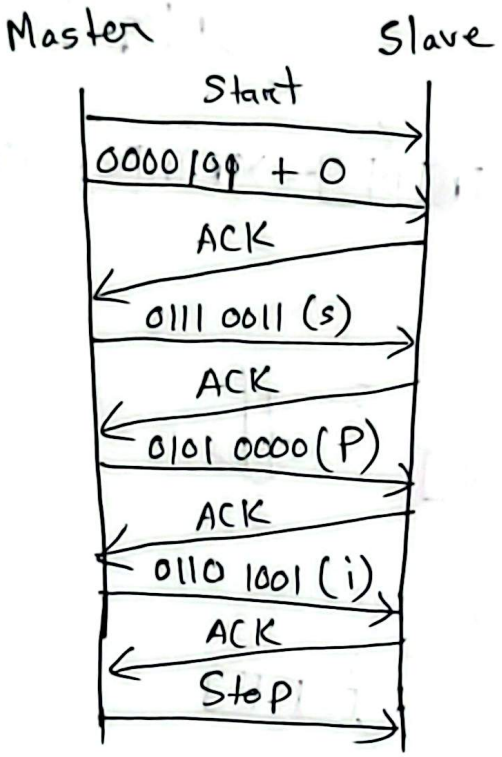
$\therefore$ 1% D.C. $\Rightarrow$ 127/100%

$\therefore$ 80% D.C. $\Rightarrow (127 \times 80\%)/(100\%)$

$= 101.6$

So, $x = 101.6$

(b) SPI is better for smaller communication system.




S.A. = 5H = 000 0101

s = 115 = 0111 0011

P = 80 = 0101 0000

i = 105 = 0110 1001

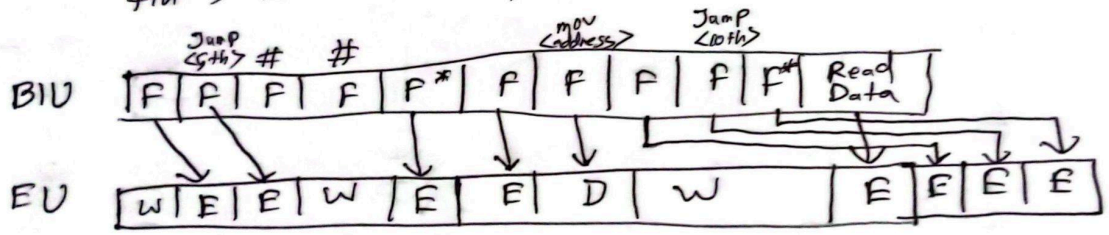
(c) Arduino would be more suitable for this task.

We are mainly using hardware devices (sensors and alert system) and doing the same task repeatedly. We do not need high processing power for this work. So, Arduino is more suitable here.

CamScanner

2nd → jump <5th> , 4th → memory read

7th → mov <address> , 9th → jump <10th>




Scanned with CamScanner

3(b) (i) $\frac{32\ KB}{4\ KB} = 8$ pages

(ii) 8 pages × 8 bytes

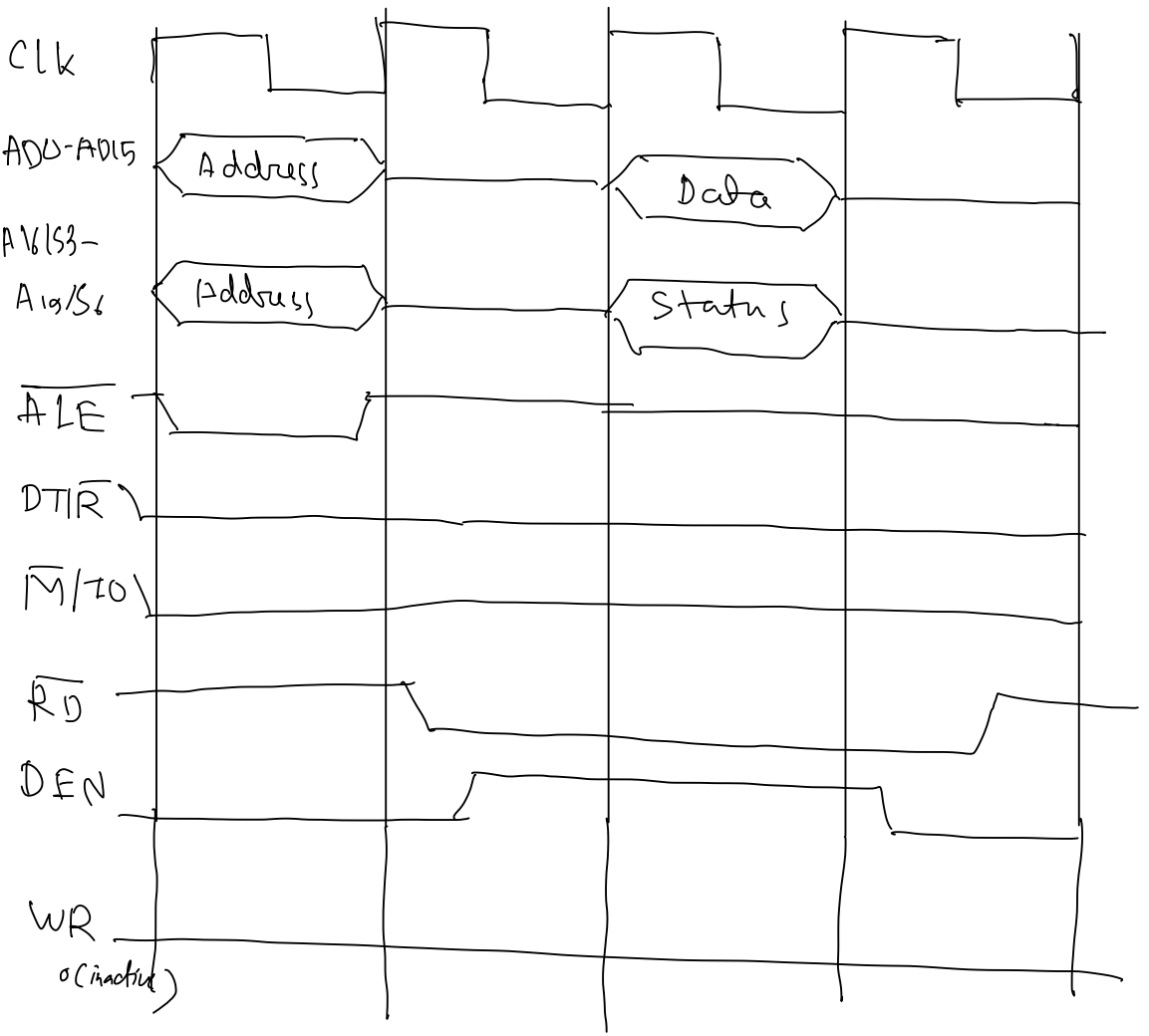
= 64 bytes

(3) Pages in primary memory = 5

∴ secondary storage = 8 − 5 = 3

∴ percentage = $\frac{3}{8} \times 100 = 37.5\%$

3(c) Read operation
Memory



4

a) expression: $-5 + 2X9A - 3^2$

$\equiv 5 - A + 9A + 9A - 3X3$

$= 5 - A + 9A + 9A - 3 - 3 - 3$

RAM content table:

| Address | Assembly | Content |
|---|---|---|
| 0H | SUB CH | 0010 1100 |
| 1H | ADD 8H | 0001 1000 |
| 2H | ADD 8H | 0001 1000 |
| 3H | SUB EH | 0010 1110 |
| 4H | SUB EH | 0010 1110 |
| 5H | SUB EH | 0010 1110 |
| 6H | OUT | 1101 XXXX |
| 7H | HLT | 1011 XXXX |

b)

| Address | Assembly | Acc = 5H | B = 10H |
|---|---|---|---|
| 0H | SUB CH | 5H − AH = −5H | AH |
| 1H | ADD 8H | −5H + 9A = 95H | 9AH |
| 2H | ADD 8H | 95H + 9AH = 2FH | 9AH |
| 3H | SUB EH | 2F − 3H = 2CH | 3H |
| 4H | SUB EH | 2C − 3H = 29H | 3H |
| 5H | SUB EH | 29H − 3H = 26H | 3H |
| 6H | OUT | 26H | 3H |
| 7H | HLT | 0H | 0H |

In 3rd instruction (at 2H), the sum is 12FH. But Acc Register holds only 8bits (from LSB).

c) $CON = L_I \overline{E_I} L_A \overline{E_A} \quad \overline{E_u} S_u L_o L_B \quad \overline{E_p} C_p \overline{L_M} \overline{CE}$

Disable $\Rightarrow$ 0 1 0 1 1 1 0 0 1 0 1 1

$= 5CBH$

i) $CON = 7EA$

$= 0\ 1\ \underset{L_A}{\boxed{1}}\ 1 \quad 1\ \underset{L_o}{\boxed{1}}\ 0 \quad 1\ 0\ 1\ \underset{\overline{CE}}{\boxed{0}}$

Operation: RAM sends the content (stored at 9H (MAR)) to Acc and Output Register.

ii) $CON = 4DB$

$= 0\ 1\ 0\ \underset{\overline{E_A}}{\boxed{0}} \quad 1\ 1\ 0\ \underset{L_B}{\boxed{1}} \quad 1\ 0\ 1\ 1$

Operation: Acc sends its content to B Register

iii) $CON = 72BH$

$= 0\ 1\ \underset{L_A}{\boxed{1}}\ 1 \quad \underset{\overline{E_u} S_u L_o}{\boxed{0\ 0\ 1}}\ 0 \quad 1\ 0\ 1\ 1$

Operation: Subtraction of Acc and B Register occurs, and result is sent to Acc and Output Regis

| T-State | Acc = 5H | = 10H B Reg | Output Reg = 7H |
|---|---|---|---|
| T4 | 5H | 10H | 5H |
| T5 | 5H | 5H | 5H |
| T6 | 5H-5H=0H | 5H | 0H |